7.8 Activities: graphs and averages

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1. Family survey

You need: squared paper, coloured pencils, family and friends (or his class, if the teacher agrees).

How: He picks a question with a few possible answers (“What’s your favourite pancake topping?” “How do you get to work or school?”), asks at least 10 people, and records the results in a tally table. Then he draws a bar chart on squared paper with a title, labelled axes and a sensible scale.

Talk about it: “What does your graph tell us? What doesn’t it tell us?” “Would the result be the same if you asked a different group?”

2. Level the towers (average)

You need: Lego or stacking blocks.

How: Build four towers of different heights, for example 3, 5, 4 and 8 blocks. Move blocks from tall towers to short towers until all four are the same height (5 each). That height is the average. Then count all the blocks (20) and share them over 4 towers (20 : 4 = 5). Same answer. Try other sets, including one where the average isn’t a whole number (3, 4, 4, 6 → 4 1/4: what happens to the leftover block?).

Talk about it: “Why does adding them up and dividing give the same answer as levelling?”

3. Weather watch (line graph and average)

You need: a thermometer outside (or a weather app), a notebook, squared paper.

How: Every day for one or two weeks, at the same time, he writes down the temperature. At the end, he draws a line graph (days along the bottom, temperature up the side) and works out the average temperature for the week.

Talk about it: “Which day was warmest? Where can you see it on the graph?” “Was any day exactly the average temperature?”

4. Pizza pie chart

You need: a paper plate or a circle drawn with a plate, coloured pencils.

How: Ask 8 (or 12) family members or friends about their favourite pizza. Split the circle into 8 (or 12) equal slices and colour one slice per person, grouping the same answers together. You’ve made a pie chart. Label each part with a fraction (3/8) and, if he likes, a percentage.

Talk about it: “Why does a pie chart work well here? Would a line graph make sense for this?”

5. Sneaky graph hunt

You need: newspapers, advertising folders, news websites with graphs.

How: Look for graphs together. For each one, ask the checklist: Is there a title? What are the axes? Does the vertical axis start at 0? Are the steps equal? How many people were asked? Who made it, and what do they want you to think? Then redraw one of them “honestly” (with the axis from 0) and compare.

Talk about it: “Is the sneaky graph actually wrong, or just unfair?” “Why would someone draw a graph this way?”

6. Make a sneaky graph yourself

You need: squared paper.

How: He gets the same small data set (sales of two ice-cream flavours: 96 and 100) and draws two bar charts: one “honest” (axis from 0) and one “sneaky” (axis from 95). Show both to someone else in the family and ask them what they think the difference is.

Talk about it: “How did it feel to make the sneaky one? Now that you know how it’s done, will you spot it more easily?”

Conversation prompts for any day

- “We spent an average of € 60 a week on groceries. Does that mean we spent € 60 every week?”
- “The news says ‘1 in 4 children…’. How many children in your class would that be?”
- “If I make a graph of your height from age 0 to 10, what shape will the line have? What about from 10 to 30?”